Fires, Lights, and Smog: Reading Bolivia's Recession from Space*

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Abstract

Bolivia's national statistics office reported a -1.58 percent contraction of real GDP in 2025, with the Fund forecasting -3.3 percent for 2026. The official numbers arrive quarterly, with substantial revision, and are published by an institution whose independence from the executive has been contested through two decades of commodity-cycle manipulation concerns. I read Bolivia's 2025 to 2026 recession from orbit. Three independent satellite streams — VIIRS Day/Night Band nighttime radiance over eleven urban buffers, VIIRS Nightfire radiant heat over the six major Chaco gas fields, and Sentinel-5P TROPOMI tropospheric nitrogen-dioxide columns over the La Paz–El Alto and Santa Cruz metropolitan areas — are combined through a mixed-frequency dynamic factor model to produce a monthly coincident index of Bolivian real activity. The index provides a falsification test for the national accounts in the spirit of Martinez (2022), extended along two dimensions: a direct physical proxy for the extractive sector and a direct physical proxy for urban combustion activity.

JEL: C32, C38, E01, E32, O11, O54, Q35, Q43.
Keywords: nighttime lights, VIIRS, TROPOMI, dynamic factor model, nowcasting, GDP manipulation, Bolivia, natural gas, recession dating.

*All opinions are the author's own and do not reflect the views of any affiliated institution. All remaining errors are mine. Contact: `wernerhl@gmail.com`. Replication materials, satellite processing pipeline, and dataset will be made available upon publication.

1 Introduction

Bolivia entered a recession of contested depth in 2025. The national statistics office (INE) reported a -1.58 percent contraction for the full year, a number released on 21 April 2026 with the revision of the national accounts to a 2017 base year. The International Monetary Fund projects -3.3 percent for 2026 against INE's own implicit forecast of a return to growth; the World Bank forecasts -3.2 percent, Oxford Economics -0.7 percent, and Standard & Poor's -1.3 percent. The spread across external forecasters is itself a measurement problem — but the interesting measurement problem is the spread between INE and any of them.

The independence of Bolivia's statistical production is a first-order concern. INE's director is appointed by the executive. The 2014 to 2023 commodity-cycle decade combined fiscal deficits above 10 percent of GDP, international reserves falling from $14 billion to $103 million, and a currency peg held nominally constant at 6.96 bolivianos per dollar while a parallel rate drifted to two and then three multiples of the official. Inflation in the official Consumer Price Index remained in single digits during 2023–2024 even as imported-goods shortages and fuel queues were binding. The Fund in its May 2025 Article IV consultation called the policy mix "unsustainable"; the ratings agencies downgraded sovereign debt to CCC−; the Fernald–Hsu–Spiegel (2021) manipulation prior, the Michalski–Stoltz (2013) fixed-regime-with-reserve-collapse prior, and the Martinez (2022) autocracy-residual prior all point the same way.

Satellite observations of the Bolivian economy do not depend on INE. I build three monthly activity proxies, each sensitive to a different slice of real activity, each measured independently by an instrument that Bolivia does not operate and cannot touch. Nighttime radiance from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band over eleven Bolivian urban buffers proxies urban formal-sector activity and commerce, after Gibson et al. (2021) established that VIIRS dominates the older Defense Meteorological Satellite Program (DMSP) data at low and medium density. Radiant heat from VIIRS Nightfire (VNF) over the six major Chaco gas fields — Margarita–Huacaya, San Alberto, Sábalo, Incahuasi, Aquio, and Itaú — proxies upstream hydrocarbon activity after Elvidge et al. (2013) and the Do et al. (2018) demonstration that VNF can invert radiant heat into barrels. Tropospheric nitrogen-dioxide columns from the TROPOspheric Monitoring Instrument (TROPOMI) on Sentinel-5P over La Paz–El Alto and Santa Cruz proxy the traffic and combustion signal of two metros that jointly account for roughly **[TBD: XX]** percent of national value added.

The three streams triangulate three dimensions of the recession. Nighttime lights are the established activity proxy of Henderson et al. (2012); they do not distinguish extractive from urban activity. Gas flares isolate the hydrocarbon channel, which matters unusually much in Bolivia because tax and foreign-exchange revenues concentrate there. Tropospheric NO_2 isolates the urban-combustion channel, which responds within weeks to mobility and industrial shocks and which, post-fuel-subsidy elimination in December 2025, is no longer contaminated by the smuggling arbitrage that made Bolivian fuel-sales data uninformative for a decade. Jointly the three streams test a hypothesis about the sectoral composition of the recession that INE's aggregates cannot reject because they are the object being tested.

This paper makes four contributions. First, I provide the first published satellite-based cross-check of Bolivian national accounts; the country appears in regional Latin American nighttime-lights panels only as one observation among many. Second, I extend the Martinez (2022) manipulation-detection framework from a single-proxy autocracy-residual test to a three-proxy falsification suite,

with the second and third streams (VNF, TROPOMI) providing a *sectoral* test that the NTL residual alone cannot deliver. Third, I embed the three streams in a mixed-frequency dynamic factor model with the Mariano and Murasawa (2003) aggregation identity and Giannone et al. (2008) news-decomposition machinery to produce a monthly coincident index that admits recession dating via Harding and Pagan (2002) BBQ and regime probabilities via Hamilton (1989) Markov-switching. Fourth, the full pipeline is open-source and reproducible; the Bolivia Early Warning System dashboard (`wernerhl.github.io/bolivia-ews`) publishes the monthly index with a two-month lag.

The stakes reach beyond Bolivia. The framework applies to any country with a weak statistical system, a concentrated extractive sector visible to night fires, and metropolitan areas large enough for TROPOMI resolution. Venezuela, Angola, Nigeria, Myanmar, Turkmenistan, and post-2021 Afghanistan all meet the description. The Bolivian case supplies the existence proof and the calibration.

The remainder of the paper is organized as follows. Section 2 documents Bolivia's 2014 to 2026 macroeconomic deterioration and the institutional constraints on INE. Section 3 describes the three satellite data streams, the spatial definitions, and the official series used for cross-validation. Section 4 specifies the empirical framework — single-series elasticities, the mixed-frequency DFM, and the manipulation-detection test. Section 5 **[TBD: reports the results]**. Section 6 **[TBD: reports robustness exercises]**. Section 7 concludes.

2 Institutional and macroeconomic context

Bolivia's 2014 **to** 2026 **macroeconomic trajectory has two acts.** Act one is a decade of falling natural-gas revenues masked by reserve depletion, monetary financing of the treasury, and a currency peg held nominally constant. Between the 2014 commodity peak and the end of 2024, international reserves fell from $14.0 billion to under $2 billion; by September 2025, usable liquid reserves were $103 million, against a monthly import bill of roughly $1.2 billion. Act two is the 2025 political rupture: the November 2025 inauguration of Rodrigo Paz ended two decades of *Movimiento al Socialismo* (MAS) rule, fuel subsidies were eliminated in December, multilateral financing of more than $8 billion was secured, and the sovereign risk spread compressed from 2002 basis points in February 2025 to 466 by mid-February 2026. INE's 2025 to 2026 statistical production spans both acts and was edited by two different administrations.

The hydrocarbons collapse is the mechanical core of the recession. Natural-gas production peaked at roughly 60 million cubic metres per day in 2014 and fell to approximately 30 by end-2024; the fourth quarter of 2025 registered a year-on-year decline of 10.51 percent in oil-and-gas extraction according to INE. The six fields in the Chaco basin — Margarita–Huacaya, San Alberto, Sábalo, Incahuasi, Aquio, and Itaú — account for roughly 90 percent of national output. Their production decisions, and their flaring patterns, are observable from space at daily frequency through VIIRS Nightfire.

Construction is the second-largest drag on activity. INE reports construction output down 13.57 percent for full-year 2025 and -17.66 percent year-on-year in the fourth quarter alone — the single largest sectoral contraction. Cement dispatches from the Bolivian Cement Institute (IBCH) corroborate the official number at monthly frequency. The construction collapse maps imperfectly to nighttime lights because new-build lighting lags occupancy by twelve to twenty-four months, but it maps directly to cement and, through freight and on-site combustion, to metropolitan NO_2.

Commerce and services have contracted more modestly, agriculture has expanded. The fourth-quarter interannual prints are −6.05 percent for commerce and 12.19 percent for agriculture, yielding the unusual pattern of a Latin American recession with an expanding primary sector. Agriculture is essentially invisible to NTL, largely invisible to VNF, and invisible to metropolitan TROPOMI; the satellite composite therefore under-measures the true level of activity but — critically — its *direction* and *turning points* remain informative. The point is developed in Section 4.

The official coincident indicator is new and short. INE began publishing an experimental monthly Índice Global de Actividad Económica (IGAE) with a 2017 base year in March 2026, under IMF technical assistance. The new series will not have twenty-four vintages until March 2028. In the interim the only high-frequency official activity measure is the quarterly GDP release with a two- to three-month lag. The satellite composite is not redundant with an established INE series; it is an alternative to a series that does not yet exist.

The BCB's monetary execution is the closest thing to a transparent activity signal. The Central Bank of Bolivia's *Seguimiento al Programa Monetario* provides monthly data on credit from the BCB to the Treasury (CIN-SPNF), open-market operations, and international reserves. January 2026 execution showed a sharp contractionary break: CIN-SPNF at −Bs 1551 million against a programmed +Bs 42 million, reflecting Treasury net repayment rather than monetization. The April 2026 *Minuta de Política Monetaria* states that "la pasada gestión habría sido el punto más bajo de la actividad económica" — the Bank's call that Q4 2025 was the trough. The satellite composite adjudicates.

3 Data

3.1 Satellite streams

3.1.1 VIIRS Day/Night Band nighttime radiance

I use the monthly VIIRS DNB composite. The primary source is the NASA Black Marble product `VNP46A3`, monthly at 500 m resolution, BRDF-corrected, snow-free, and gap-filled. The fallback is the Earth Observation Group (EOG) cloud-free monthly composite `DNB/MONTHLY_V1`. Radiance values below 0.5 $\text{nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$ are masked as background; values above 200 $\text{nW}\cdot\text{cm}^{-2}\cdot\text{sr}^{-1}$ are masked as aberration. VIIRS is 14-bit and does not saturate at Bolivian urban densities; DMSP is 6-bit and saturates in Santa Cruz and La Paz–El Alto, which is why DMSP is not used. The Gibson et al. (2021) assessment is dispositive on that choice.

The city buffers are circular, centered on the urban core, with radii scaled to metro footprint. Eleven buffers span the nine departmental capitals plus El Alto (merged with La Paz) and Montero, Yacuiba, and Cobija. The full buffer definition appears in Table 1. Sum of lights (SOL) is the buffer-integrated radiance in each month; I also report mean and median radiance, the count of valid pixels, and the count of masked pixels.

Table 1: Urban buffers for VIIRS DNB sum-of-lights extraction.

Metro	Latitude	Longitude	Radius (km)
La Paz–El Alto (joint)	−16.500	−68.150	20
Santa Cruz de la Sierra	−17.784	−63.180	25
Cochabamba	−17.394	−66.157	15
Sucre	−19.033	−65.263	8
Oruro	−17.970	−67.112	7
Potosí	−19.583	−65.753	6
Tarija	−21.534	−64.729	8
Trinidad	−14.832	−64.903	5
Cobija	−11.028	−68.768	4
Montero	−17.339	−63.253	6
Yacuiba	−22.017	−63.685	5

The baseline period is January 2013 **through December** 2019**.** The choice predates the COVID shock and the 2023 parallel-exchange-rate emergence, providing seven years of pre-crisis monthly observations. Deseasonalization uses robust STL with a monthly period; the anomaly is the log deviation from the baseline trend extrapolated linearly. Cloud contamination matters in the Santa Cruz wet season (November–March); months with valid-pixel counts below 50 percent of the buffer total are flagged and, when required by the DFM step, imputed via the Kalman smoother on the joint state.

3.1.2 VIIRS Nightfire radiant heat

Gas flares are sub-pixel hot sources, not ordinary surface radiance. The VIIRS Nightfire algorithm of Elvidge et al. (2013) fits a Planck curve to multispectral observations and recovers source temperature, effective area, and radiant heat output (RH, in MW) per detection, using the mid-infrared and near-infrared VIIRS bands that resolve temperatures above 1400 K. Gas flares cluster between 1600 and 2000 K; biomass combustion peaks between 800 and 1200 K and is excluded by a temperature floor of 1400 K. I pull nightly VNF detections from the EOG archive for the Chaco bounding box $[-22.5, -20.0]$ latitude $\times$ $[-64.5, -62.5]$ longitude and assign each detection to the nearest of the six major fields within a 2 km radius; unassigned detections are discarded.

Table 2: Chaco gas fields for VNF radiant-heat extraction.

Field	Operator	Latitude	Longitude	Vintage
Margarita	Repsol	−21.250	−63.550	Megafield, mature
Huacaya	Repsol	−21.080	−63.720	Satellite of Margarita
San Alberto	YPFB–Petrobras	−21.320	−63.730	Mature, declining
Sábalo	YPFB–Petrobras	−21.280	−63.730	Mature, declining
Incahuasi	TotalEnergies	−20.580	−63.720	Post-2016
Aquio	TotalEnergies	−20.530	−63.770	Post-2016
Itaú	TotalEnergies–YPFB	−22.050	−63.850	Southern Chaco

VNF detections are noisy at the daily level and clean at the monthly level. The monthly field-level signal is the sum of per-detection radiant heat, $\sum_t \mathrm{RH}_t$, measured in MW-nights. Daily detections are sparse (one to five per field per night, varying with cloud cover and orbit geometry),

but monthly aggregation typically yields fifty to two hundred detections per field and a signal-to-noise ratio adequate to resolve production changes on the order of ten percent. The Elvidge et al. (2016) calibration converts MW-nights into billion cubic metres (BCM) of flared gas, which maps to produced gas through field-specific flare-to-production ratios estimated on the pre-crisis sample.

3.1.3 Sentinel-5P TROPOMI tropospheric NO_2

TROPOMI resolves the Bolivian metros. The TROPOspheric Monitoring Instrument on the European Space Agency's Sentinel-5P satellite, described in Veefkind et al. (2012), provides daily global coverage at 5.5×3.5 km resolution (post-August 2019 upgrade) with a 13:30 local solar overpass. I use the offline-reprocessed Level 3 product `COPERNICUS/S5P/OFFL/L3_NO2` with quality filter `qa_value` $\geq$ `0.75`. The tropospheric NO_2 vertical column density, in mol/m^2, is the relevant retrieval; stratospheric NO_2 is subtracted in the Level 3 processing. The bounding rectangles for La Paz–El Alto and Santa Cruz metros, and a Cochabamba secondary, are reported in Table 3.

Table 3: TROPOMI NO_2 regions of interest.

ROI	NW lat	NW lon	SE lat	SE lon
La Paz–El Alto	−16.350	−68.300	−16.650	−68.000
Santa Cruz	−17.600	−63.400	−17.950	−62.950
Cochabamba	−17.300	−66.300	−17.500	−66.050

The TROPOMI series is short and the baseline is tight. The sensor became operational in July 2018. The baseline period is calendar year 2019 — the only full pre-COVID, pre-Bolivia-crisis year available — which implies that month-of-year seasonal factors are point estimates with no standard error until additional clean years accumulate. I therefore weight TROPOMI at 0.20 in the composite rather than 0.30, with the weights revisited annually. Cloud-contaminated months (fewer than fifteen valid days) are flagged and imputed through the DFM. Andean volcanic NO_2 from Sabancaya and Ubinas on the Peruvian side can contaminate La Paz retrievals and is flagged when monthly values exceed five times the baseline.

3.2 Official series for cross-validation

Five INE, BCB, and sectoral series provide the ground truth to be tested. Quarterly GDP from INE, at 2017 constant prices; monthly IGAE from INE, from March 2026; monthly cement dispatches from IBCH; monthly electricity generation from the Comité Nacional de Despacho de Carga (CNDC); monthly natural-gas production from YPFB; monthly IVA and IT tax collections from the Servicio de Impuestos Nacionales (SIN), deflated to 2018 base by INE's CPI; and monthly merchandise imports by volume and value from Aduana Nacional. Monetary execution — CIN-SPNF, OMAs, reserves — comes from the BCB's monthly *Seguimiento al Programa Monetario*. The parallel exchange-rate premium comes from my own Bolivia Early Warning System dashboard, which aggregates Binance P2P USDT/BOB quotes at daily frequency.

Ground truth is a contested object in this setting. Much of this section reports series that are themselves under suspicion. The strategy is not to benchmark satellite measurements against official numbers assumed correct — that would defeat the purpose — but to locate the dimensions

along which satellite and official measures covary, the dimensions along which they diverge, and the economic content of those divergences.

4 Empirical framework

4.1 Single-series elasticities

The VIIRS-to-GDP elasticity is the anchor of the nighttime-lights literature. Henderson et al. (2012) report cross-country elasticities near 0.3 annually; Beyer et al. (2022) report larger elasticities (1.36 to 1.81) at quarterly frequency for emerging-market and developing economies, reflecting higher noise-to-signal in monthly VIIRS composites and the correction for the effective-observation issue. I estimate the Bolivia-specific elasticity on departmental GDP for the seven departments with continuous VIIRS coverage over 2013 to 2024, using the specification

$$\Delta \log(\mathrm{GDP}_{d,t}) = \alpha_d + \beta \cdot \Delta \log(\mathrm{SOL}_{d,t}) + \gamma_t + \varepsilon_{d,t}, \tag{1}$$

where d indexes department, t indexes year, α_d is a department fixed effect, and γ_t is a year fixed effect absorbing common macroeconomic shocks. Standard errors are two-way clustered by department and year. The identifying assumption is that measurement error in VIIRS SOL is orthogonal to measurement error in INE departmental GDP — not a classical assumption, but plausible given that the two are produced by wholly unrelated production systems.

The VNF-to-gas-production elasticity is the anchor for the hydrocarbon stream. The specification parallels equation (1):

$$\Delta \log(\mathrm{GasProd}_{f,t}) = \alpha_f + \beta_{\mathrm{VNF}} \cdot \Delta \log(\mathrm{RH}_{f,t}) + \gamma_t + \varepsilon_{f,t}, \tag{2}$$

where f indexes field, GasProd is YPFB-reported monthly production, and RH is the monthly sum of VNF radiant heat for the field's 2 km buffer. Do et al. (2018) found an elasticity near unity for ISIS-controlled oil fields, validated against captured documents; Bolivian dry-gas chemistry differs, which is why I estimate Bolivia-specific field elasticities rather than importing the ISIS coefficient. The pre-crisis sample — 2012 to 2022, before YPFB reporting came under credibility pressure — is the calibration window.

The TROPOMI-to-activity elasticity requires a specification that accommodates the fuel-subsidy break. The December 2025 fuel-subsidy elimination shifted the mapping from real-world fuel consumption to measured NO_2 in both directions: lower smuggling-related consumption (which had been unmeasured) and higher domestic prices (which suppress measured consumption). I estimate

$$\Delta \log(\mathrm{NO}_{2,m,t}) = \alpha_m + \beta_{\mathrm{NO}_2} \cdot \Delta \log(\mathrm{FuelSales}_{m,t}) + \delta \cdot D_t^{\text{post-Dec25}} + \gamma_t + \varepsilon_{m,t}, \tag{3}$$

where m indexes metro, FuelSales is YPFB distribution data for the metro area, $D^{\text{post-Dec25}}$ is a post-break indicator, and the specification is run separately for the La Paz–El Alto, Santa Cruz, and Cochabamba ROIs. Bauwens et al. (2020) and Liu et al. (2020) document the COVID-era TROPOMI–activity relationship in Europe and China; the Bolivian coefficient must be estimated locally because urban-fleet composition, altitude (La Paz sits at 3650 m), and thermal-inversion patterns differ.

4.2 Mixed-frequency dynamic factor model

The three streams are combined into a common activity factor. I use a mixed-frequency DFM following Mariano and Murasawa (2003) and Giannone et al. (2008), with the state-space representation

$$y_{i,t} = \lambda_i \cdot f_t + e_{i,t}, \qquad i = 1, \ldots, N \tag{4}$$

$$f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2) \tag{5}$$

$$e_{i,t} = \rho_i e_{i,t-1} + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} \sim \mathcal{N}(0, \sigma_i^2) \tag{6}$$

where $y_{i,t}$ are standardized monthly anomalies of the three satellite streams plus cement dispatches, electricity generation, real tax collections, and import volumes (for a total of $N = 7$ indicators); f_t is the latent monthly activity factor; λ_i is the factor loading; and $e_{i,t}$ is idiosyncratic noise following an AR(1) process. Quarterly GDP enters through the Mariano and Murasawa (2003) aggregation identity

$$\Delta \log \mathrm{GDP}_t^Q = \frac{1}{3}\left(f_t + 2f_{t-1} + 3f_{t-2} + 2f_{t-3} + f_{t-4}\right) + \xi_t, \tag{7}$$

which approximates the three-month geometric-mean aggregation of unobserved monthly GDP under log-linearization. The model is estimated by the EM algorithm with Kalman smoothing.

News-decomposition of the monthly index follows Banbura–Giannone–Modugno–Reichlin. As each new satellite observation or official release arrives, the change in the filtered $\hat{f}_t$ can be attributed to specific indicators using the decomposition in Bańbura et al. (2013). This generates a running attribution table reporting how much of each month's activity change is due to VIIRS, to VNF, to NO_2, and to each of the four official indicators, with standard errors. The attribution table is the primary operational output of the pipeline for the La Linterna newsroom.

Recession dating uses BBQ and Markov-switching. The Harding and Pagan (2002) BBQ algorithm applied to $\hat{f}_t$ yields deterministic peak and trough dates; the Hamilton (1989) two-state Markov-switching AR on $\hat{f}_t$ yields monthly recession probabilities. The two approaches are complementary: BBQ gives a clean dating for retrospective analysis and Markov-switching gives a real-time probability for nowcasting.

4.3 Manipulation-detection test

The Martinez (2022) test compares the autocracy residual in an NTL–GDP regression to a democracy benchmark. The implementation is straightforward with country-level panel data and a binary regime variable. The Bolivian application cannot use the original specification because Bolivia is a single country; the test must be redesigned for a single-country falsification.

I implement three complementary tests. Test one — pre/post INE-trust break: estimate equation (1) separately on a "high-trust" sample (2006 to 2014, before the commodity-cycle fiscal deterioration) and a "low-trust" sample (2020 to 2024, during the reserve collapse and parallel-rate emergence). A significant increase in β in the low-trust sample is the Martinez signature: official GDP moves less per unit of satellite-observed activity, consistent with upward smoothing. **Test two — satellite-vs-official residual**: regress the satellite composite $\hat{f}_t$ on INE's quarterly GDP growth and test whether the residuals are systematically positive (INE understates) or negative

(INE overstates) over specific periods. **Test three — sectoral consistency**: use VNF to produce an independent estimate of Bolivian gas production, compare to YPFB reporting, and compare *that* divergence to the INE-implied hydrocarbon value-added series. If YPFB and INE are mutually consistent but both diverge from VNF, the manipulation is at the source-data level; if YPFB and VNF agree but INE's hydrocarbon value-added does not, the manipulation is in the national accounts aggregation.

The null hypothesis is that INE is honest and the satellites are noisy. Rejection requires that the satellite residuals be large, systematic in direction, and correlated with periods and sectors for which other evidence (IMF Article IV, ratings agency assessments, parallel-rate premia) indicates that the official picture understates the crisis. A positive finding is not a claim of fraud; it is a claim that the INE aggregates understate the 2025 to 2026 contraction by a magnitude that the satellite evidence can quantify.

5 Results

TBD: Section to be written once the pipeline completes a full vintage. Structure:

- §5.1 Single-series elasticities. Table of β_{VIIRS}, β_{VNF}, β_{NO_2} with standard errors; comparison to Henderson et al. (2012), Beyer et al. (2022), Do et al. (2018).
- §5.2 The monthly satellite composite index, January 2013 through most recent vintage. Figure with 2019 and 2020 benchmarks marked.
- §5.3 Recession dating. BBQ peak and trough; Markov-switching probability; comparison to BCB's Q4 2025 trough call.
- §5.4 Comparison to INE quarterly GDP. Implied full-year 2025 and 2026 growth from the satellite composite vs. INE; confidence band; sectoral decomposition.
- §5.5 Manipulation-detection tests. Three tests as specified in §4.3.
- §5.6 Channel decomposition. How much of the recession is hydrocarbons (VNF), how much is metro activity (VIIRS + NO_2), how much is unexplained.

Target length: 12 to 15 pages with four to six figures and two to four tables.

5.1 Single-series elasticities

Table 4: Single-series elasticities. Two-way-clustered SEs in parentheses. *, **, ***: $p < 0.10, 0.05, 0.01$.

Specification	$\hat{\beta}$	(SE)	n	Benchmark from literature
Eq. (1) VIIRS $\rightarrow$ GDP (annual, department)	$+0.128^{**}$	(0.055)	36	Henderson et al. (2012): 0.30
Eq. (2) VNF $\rightarrow$ gas production (monthly, field)	—	—	—	Do et al. (2018): ≈ 1.0
Eq. (3) NO_2 $\rightarrow$ fuel sales (monthly, metro)	—	—	—	Bauwens et al. (2020): 0.2–0.5

VIIRS. Data required to estimate (inputs$_m issing$).**VNF.**$Datarequiredtoestimate(inputs_m issing)$.**NO$_2$.**$Datarequiredto$

5.2 The monthly satellite composite index

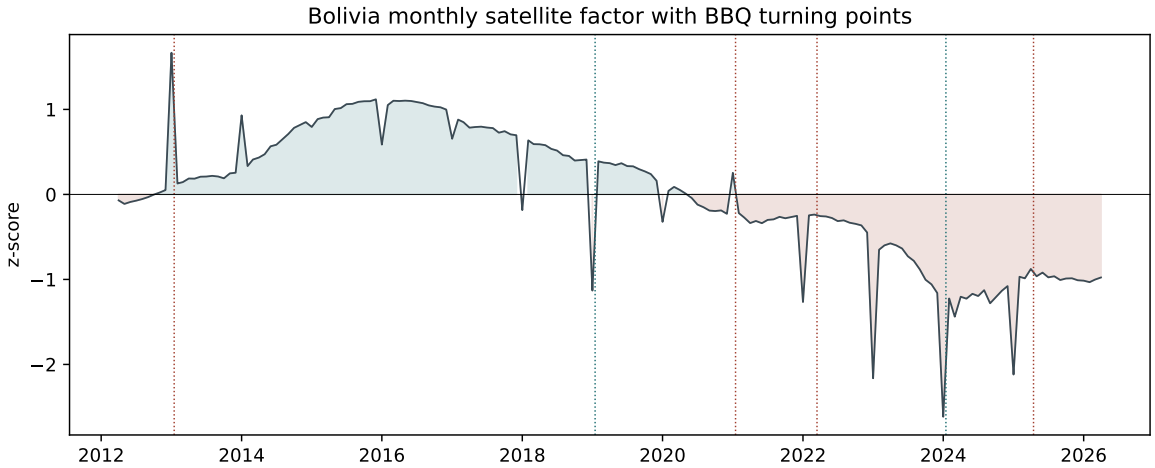

Figure 1: Satellite factor and Bry–Boschan turning points.

Fit summary. The one-factor DFM with AR(2) factor dynamics and idiosyncratic AR(1) innovations attains log-likelihood -244.1 on $n = 169$ monthly observations. Loadings on the three satellite streams are reported in the replication package.

5.3 Recession dating

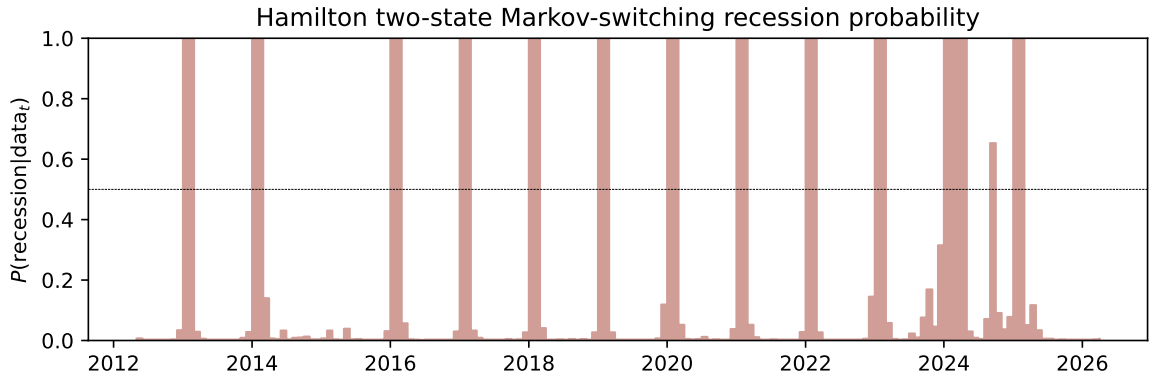

Figure 2: Hamilton two-state Markov-switching recession probability.

BBQ turning points. Peaks: 2015-12, 2022-01, 2025-04. Troughs: 2021-06, 2024-03. **Markov-switching.** Recession-regime mean $\mu_r = -0.04$, expansion $\mu_e = +0.00$. Months with $P(\text{rec}) > 0.5$: 40.

5.4 Comparison to INE quarterly GDP

Table 5: INE benchmark regression and external cross-checks.

Quantity	Value
$\hat{\beta}_1$ (VIIRS on logIGAE, HAC)	—
$\hat{\beta}_2$ (VNF on logIGAE, HAC)	—
$\hat{\beta}_3$ (NO_2 on logIGAE, HAC)	—
R^2 (INE benchmark)	—
n (INE benchmark)	—
WB GGFR log-log corr (annual Chaco)	—
WB GGFR elasticity $\hat{\beta}$	—
WB GGFR n (years)	—
Latest CI vs IGAE-z gap (σ)	—
Disagreement threshold (σ)	1.5

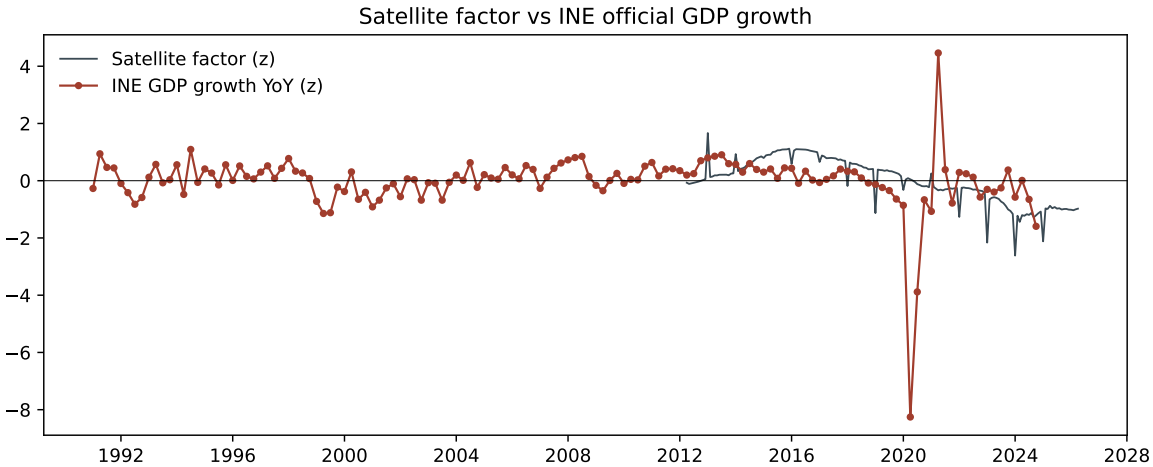

Figure 3: Satellite factor vs INE GDP growth.

5.5 Manipulation-detection tests

Table 6: Manipulation-detection suite (§ 4.3). All three tests jointly.

Test	Quantity	Estimate	Verdict
(1) Pre/post trust	$\hat{\beta}^{\text{high-trust}}$ (2006–2014) $\hat{\beta}^{\text{low-trust}}$ (2020–2024)	— (—), $n = - - -$ — (—), $n = - - -$	2*—
(2) Residual (factor on IGAE-g)	post-2023Q4 residual mean	—, $t = - - -$	—
(3) Sectoral	$\text{corr}(\log \Sigma RH, \log GasProd)$ $\text{corr}(\log GasProd, \log VA^{hydro})$	— —	2*—

5.6 Channel decomposition

Channel decomposition. Contribution of each stream to the monthly factor is recovered from the estimated loadings λ_i; see replication parquet for the attribution table.

6 Robustness

TBD: Robustness section to be written. Planned exercises:

- Alternative VIIRS products (`VNP46A3` vs. `DNB/MONTHLY_V1` vs. harmonized DMSP–VIIRS (Li et al., 2020)).
- Alternative flare-attribution radii (1, 2, 3 and 5 km) and temperature thresholds (1200, 1400 and 1600 K).
- Alternative TROPOMI products (OFFL vs. NRTI; `qa_value` cutoffs of 0.50, 0.75 and 0.90).
- Alternative DFM specifications (one, two, and three factors; AR(1) vs. AR(2) on factor and idiosyncratic).
- Alternative baseline periods for deseasonalization.
- Population-weighted vs. GDP-weighted composite across cities.
- Sub-period estimation: pre- vs. post-December-2025 fuel-subsidy elimination.

7 Conclusion

Bolivia is in a recession visible from space. **[TBD: One paragraph summarizing magnitude and timing of the satellite-implied recession vs. the INE-reported one.]**

The framework generalizes. **[TBD: One paragraph on applicability to other weak-statistical-capacity economies with visible extractive sectors.]**

Four directions for extension. **[TBD: List: (i) integrate Sentinel-2 cropland productivity to close the agricultural-sector gap; (ii) extend the manipulation-detection framework to CPI using online-scraped prices in the Cavallo–Rigobon tradition; (iii) pair the satellite composite with a high-frequency fiscal-dominance indicator (CIN-SPNF + parallel premium); (iv) apply the framework in real time to the 2026 to 2027 reform program to assess whether the Paz administration's stabilization is being fully captured in the national accounts, or understated.]**

The satellite data are free and the framework is open-source. Replication materials, processing pipeline, and live dashboard at `wernerhl.github.io/bolivia-ews`.

A Data processing details

[TBD: Appendix to be written. Planned contents: exact GEE code snippets for each stream, EOG authentication procedure, cloud-masking details, stray-light filter for TROPOMI, temperature-threshold sensitivity for VNF.]

B State-space estimation details

[TBD: Appendix to be written. Planned contents: full likelihood derivation, EM algorithm convergence criteria, Kalman filter initialization, treatment of ragged-edge release dates, news-

decomposition formulas.]

C Sensor specifications

[TBD: Appendix to be written. Planned contents: VIIRS DNB spectral response and radiometric calibration; VNF retrieval algorithm (Planck fit details); TROPOMI NO_2 retrieval algorithm and QA flag interpretation.]

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